Project Report

Robust Beamforming for RIS-Assisted Wireless Networked Control Systems

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1. Introduction and Paper Summary

Introduction

A Reconfigurable Intelligent Surface is a planar array of passive, programmable reflective elements whose phase shifts can be tuned to reshape the radio propagation environment, thus strengthening the desired signal paths and suppressing interference without the power overhead of conventional active relays. When a multi-antenna controller (treated as a transmitter in this case) is paired with an optimally configured RIS, the joint active beamforming (at the controller) and passive beamforming (at the RIS) can substantially reduce the transmit power required to maintain an acceptable link quality for all Production Devices, or PDs (treated as a receiver in this case) in the Wireless Networked Control System (WNCS).

Paper Summary

The paper by Biao Ma et al. (IEEE TVT, 2026) addresses this setting. Their key contributions are:

1. The control-stability constraint is converted to an outage-probability constraint $\Pr\{\gamma_m \leq \gamma_{\text{th}}\} \leq 1 - \xi_m^*$, where ξ_m^* the lower bound on the probability that the received SNR exceeds the threshold,, equivalently imposing an upper bound $1 - \xi_m^*$ on the outage probability.

2. The resulting non-convex problem is decomposed into two convex sub-problems solved iteratively, wherein one is for the active beamformer w_b and the other is for the passive RIS phase vector $\mathbf{v}$.

3. Channel uncertainty (imperfect CSI) is handled by converting probabilistic constraints via the Bernstein-Type Inequality, then linearised around a feasible point using first-order Taylor approximation to eliminate the non-convex rank-1 constraint.

This project implements the full algorithm in MATLAB using the standard convex optimization software package CVX.

2. System Model

Network Description

The system (Fig. 1 of the paper) consists of:

- A Controller with a Uniform Planar Array (UPA) of $N_b = N_{b1} \times N_{b2}$ antennas ($4 \times 4 = 16$ in our simulations).
- An RIS with $N_r = N_{r1} \times N_{r2}$ passive reflective elements ($4 \times 4 = 16$).
- $M = 5$ Production Devices (PDs), each a single-antenna receiver.

The controller broadcasts a common control signal $s_{b,n}$ (with $\mathbb{E}[|s_{b,n}|^2] = 1$) to all PDs at discrete time n. The received signal at PD m is

$$y_{m,n} = \left(\mathbf{h}_{m,n}^H \mathbf{\Phi}_n \mathbf{H} + \mathbf{g}_{m,n}^H\right) \mathbf{w}_{b,n}\, s_{b,n} + \varpi_{m,n}, \tag{2.1}$$

where:

- $\mathbf{\Phi}_n = \mathrm{diag}(e^{j\phi_1}, \ldots, e^{j\phi_{N_r}}) \in \mathbb{C}^{N_r \times N_r}$ is the RIS phase matrix,
- $\mathbf{H} \in \mathbb{C}^{N_r \times N_b}$ is the controller-to-RIS channel matrix,
- $\mathbf{g}_{m,n} \in \mathbb{C}^{N_b \times 1}$ is the direct controller-to-mth PD channel,
- $\mathbf{h}_{m,n} \in \mathbb{C}^{N_r \times 1}$ is the RIS-to-mth PD channel,
- $\mathbf{w}_{b,n} \in \mathbb{C}^{N_b \times 1}$ is the active beamforming vector,
- $\varpi_{m,n} \sim \mathcal{CN}(0, \sigma_m^2)$ is additive white Gaussian noise (AWGN).

The output Signal to Noise Ratio (SNR) at the mth PD is therefore

$$\gamma_{m,n} = \frac{\left|\left(\mathbf{h}_{m,n}^H \mathbf{\Phi}_n \mathbf{H} + \mathbf{g}_{m,n}^H\right) \mathbf{w}_{b,n}\right|^2}{\sigma_m^2}. \tag{2.2}$$

Imperfect CSI Model

Because the controller and RIS are stationary, $\mathbf{H}$ can be accurately measured (assumption from the paper). The PD channels, however, are subject to estimation and quantisation errors given by:

$$\mathbf{g}_i = \tilde{\mathbf{g}}_i + \Delta\mathbf{g}_i, \quad \Delta\mathbf{g}_i \sim \mathcal{CN}(\mathbf{0},\, \sigma_g^2\, \mathbf{I}_{N_b}), \tag{2.3}$$

$$\mathbf{h}_i = \tilde{\mathbf{h}}_i + \Delta\mathbf{h}_i, \quad \Delta\mathbf{h}_i \sim \mathcal{CN}(\mathbf{0},\, \sigma_h^2\, \mathbf{I}_{N_r}), \tag{2.4}$$

where $\tilde{\mathbf{g}}_i, \tilde{\mathbf{h}}_i$ are the estimated channels, and $\Delta\mathbf{g}_i, \Delta\mathbf{h}_i$ are zero-mean complex Gaussian error vectors. In our implementation we set $\sigma_g = \sigma_h = 0.01$.

Control Model

Each PD is governed by a linear time-invariant discrete-time equation:

$$\mathbf{x}_{m,n+1} = \mathbf{\Omega}_m \, \mathbf{x}_{m,n} + \mathbf{\Lambda}_m \, \mathbf{u}_{m,n} + \mathbf{n}_{m,n}, \tag{2.5}$$

where $\mathbf{x}_{m,n}$ is the state vector, $\mathbf{u}_{m,n} = -\mathbf{K}_m \mathbf{x}_{m,n}$ is the control input gain, and $\mathbf{n}_{m,n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_m)$ is process noise. The shared physical parameters for all $M = 5$ PDs are

$$\mathbf{\Omega} = \begin{bmatrix} 0.37 & 0 \\ 0.67 & 1 \end{bmatrix}, \quad \mathbf{\Lambda} = \begin{bmatrix} 0.67 \\ 0.37 \end{bmatrix}, \quad \mathbf{R}_m = \mathbf{I}_{N_{\text{st}}}. \tag{2.6}$$

When the SNR at PD_m is above threshold γ_{th}, we consider it to be a closed-loop system ($\mathbf{\Omega}_{m1} = \mathbf{\Omega}_m + \mathbf{\Lambda}_m \mathbf{K}_m$); the system is considered to be open-loop ($\mathbf{\Omega}_{m0} = \mathbf{\Omega}_m$)

$$\mathbf{x}_{m,n+1} = \begin{cases} \mathbf{\Omega}_{m1} \mathbf{x}_{m,n} + \mathbf{n}_{m,n}, & \gamma_{m,n} \geq \gamma_{\text{th}}, \\ \mathbf{\Omega}_{m0} \mathbf{x}_{m,n} + \mathbf{n}_{m,n}, & \gamma_{m,n} < \gamma_{\text{th}}. \end{cases} \tag{2.7}$$

3. Communication & Control Co-Design

Lyapunov Stability Condition

To measure control performance, a Lyapunov-like function is defined per PD:

$$\Xi_m(\mathbf{x}_m) = \mathbf{x}_m^\top Q_m \mathbf{x}_m; \quad Q_m \succ 0. \tag{3.1}$$

The control stability condition requires this function to decrease in expectation at rate $\rho_m < 1$:

$$\mathbb{E}\left[\Xi_m(\mathbf{x}_{m,n+1}) \,\middle|\, \mathbf{x}_{m,n}\right] \leq \rho_m \, \Xi_m(\mathbf{x}_{m,n}) + \mathrm{Tr}(Q_m R_m). \tag{3.2}$$

Expanding this expectation using the open/closed-loop conditions and rearranging yields the communication reliability requirement:

$$\Pr\{\gamma_{m,n} \geq \gamma_{\mathrm{th}}\} \geq \xi_m^*, \tag{3.3}$$

where ξ_m^* is the supremum of the right-hand side of

$$\xi_m = \sup_{z \neq 0} \frac{z^\top (\mathbf{\Omega}_{m0}^\top Q_m \mathbf{\Omega}_{m0} - \rho_m Q_m) z}{z^\top (\mathbf{\Omega}_{m0}^\top Q_m \mathbf{\Omega}_{m0} - \mathbf{\Omega}_{m1}^\top Q_m \mathbf{\Omega}_{m1}) z}. \tag{3.4}$$

This is solved by setting $Z = zz^\top$ and lifting to $U \succeq 0$, we get a formulation that is solvable by CVX.

Joint Optimisation Problem

The original problem minimises transmit power subject to control stability and unit-modulus RIS constraints (Eq. (18) in the paper):

$$\min_{\mathbf{w}_b, \mathbf{\Phi}} \|\mathbf{w}_b\|^2 \quad \text{s.t.} \quad \text{C1: } \Pr\{\gamma_m \leq \gamma_{\mathrm{th}}\} \leq 1 - \xi_m^*, \ \forall m; \quad \text{C2: } |[\mathbf{\Phi}]_{n_r n_r}| = 1, \ \forall n_r. \tag{3.5}$$

This problem is non-convex due to: the probabilistic constraint C1 under uncertain CSI, the unit-modulus constraint C2, and the coupling of $\mathbf{w}_b$ and $\mathbf{\Phi}$ in the SNR expression.

4. Proposed Algorithm

Breakdown of the Problem and the Solution

The joint problem is split into two sub-problems solved alternately until convergence.

Sub-problem 1: Active Beamforming (fixed $\mathbf{\Phi}$)

With $\mathbf{\Phi}$ fixed, we define $\mathbf{D} = [\mathbf{\Phi}\mathbf{H}, \mathbf{I}_{N_b}]^T$ and $\mathbf{b}_m = [\tilde{\mathbf{h}}_m + \Delta\mathbf{h}_m;\ \tilde{\mathbf{g}}_m + \Delta\mathbf{g}_m]$. The SNR constraint becomes:

$$\Pr\left\{|\mathbf{b}_m^H \mathbf{D}\mathbf{w}_b|^2 \leq \gamma_{\text{th}}\sigma_m^2\right\} \leq 1 - \xi_m^*, \quad \forall m. \tag{4.1}$$

Substituting the error term $\Delta\mathbf{b}_m = \mathbf{E}_b^{1/2}\mathbf{x}_{b,m}$ with $\mathbf{x}_{b,m} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$, the constraint takes the form (Eq. (22) in the paper):

$$\Pr\left\{\mathbf{x}_{b,m}^H \mathbf{A}_{b,m}\mathbf{x}_{b,m} + 2\,\mathrm{Re}(\mathbf{x}_{b,m}^H \boldsymbol{\alpha}_{b,m}) \leq \delta_{b,m}\right\} \leq 1 - \xi_m^*. \tag{4.2}$$

Bernstein-Type Inequality

The Bernstein inequality converts this probabilistic constraint into a manageable deterministic form. With $\sigma_{b,m} = -\ln(1 - \xi_m^*)$, the following requirement holds:

$$\min_{\mathbf{W}_b \succeq \mathbf{0},\, \mathbf{w}_b} \quad \mathrm{Tr}(\mathbf{W}_b) \tag{4.3}$$

$$\text{s.t.} \quad \mathrm{Tr}(\mathbf{A}_{b,m}) - \sqrt{2\sigma_{b,m}}\,\mu_{b,m} - \sigma_{b,m}v_{b,m} - \delta_{b,m} \geq 0, \quad \forall m, \tag{4.4}$$

$$\left\| \begin{bmatrix} \mathrm{vec}(\mathbf{A}_{b,m}) \\ \sqrt{2}\,\alpha_{b,m} \end{bmatrix} \right\| \leq \mu_{b,m}, \quad \forall m, \tag{4.5}$$

$$v_{b,m}\mathbf{I} + \mathbf{A}_{b,m} \succeq \mathbf{0}, \quad v_{b,m} \geq 0, \quad \forall m, \tag{4.6}$$

$$\mathrm{rank}(\mathbf{W}_b) = 1. \tag{4.7}$$

First-Order Convex Approximation (Rank-1 Relaxation)

The non-convex rank-1 constraint $\mathbf{W}_b = \mathbf{w}_b\mathbf{w}_b^H$ is linearised around the current iterate $\mathbf{w}_b^{(t)}$:

$$\mathbf{W}_b \approx \mathbf{w}_b^{(t)}\mathbf{w}_b^H + \mathbf{w}_b\,\mathbf{w}_b^{(t)H} - \mathbf{w}_b^{(t)}\mathbf{w}_b^{(t)H}. \tag{4.8}$$

Sub-problem 2: Passive Beamforming / RIS Phase (fixed $\mathbf{w}_b$)

With $\mathbf{w}_b$ fixed, define $\tilde{\mathbf{v}}^H = [\mathbf{v}^H \operatorname{diag}(\mathbf{H}\mathbf{w}_b)\ \mathbf{w}_b^\top]$ and $\mathbf{c}_m = [\tilde{\mathbf{h}}_m^*; \tilde{\mathbf{g}}_m^*]$. The constraint becomes $\Pr\{|\tilde{\mathbf{v}}^H \mathbf{c}_m|^2 \le \gamma_{\text{th}}\sigma_m^2\} \le 1 - \xi_m^*$. Applying the same Bernstein machinery and first-order linearisation around $\mathbf{v}^{(t)}$ yields a similar problem statement with the objective:

$$\min_{\mathbf{v}} \quad 1 \tag{4.9}$$

$$\text{s.t.} \quad \text{C1:}\ \Pr\left\{\left|\tilde{\mathbf{v}}^H \mathbf{c}_m\right|^2 \le \gamma_{th}\sigma_m^2\right\} \le 1 - \xi_m^*, \tag{4.10}$$

$$\text{C2:}\ |[\mathbf{v}]_{n_r}| = 1, \quad \forall n_r. \tag{4.11}$$

Similar to the previous procedure, the outage probability constraint C1 can be expressed as:

$$\operatorname{Tr}(\mathbf{B}_{c,m}) - \sqrt{2\sigma_{c,m}}\,\mu_{c,m} - \sigma_{c,m}v_{c,m} - \delta'_{c,m} \ge 0, \quad \forall m, \tag{4.12}$$

$$\left\| \begin{bmatrix} \operatorname{vec}(\mathbf{B}_{c,m}) \\ \sqrt{2}\,\beta_{c,m} \end{bmatrix} \right\| \le \mu_{c,m}, \quad \forall m, \tag{4.13}$$

$$v_{c,m}\mathbf{I} + \mathbf{B}_{c,m} \succeq \mathbf{0}, \quad v_{c,m} \ge 0, \quad \forall m. \tag{4.14}$$

Algorithm Summary

Input: $\{\mathbf{H}, \mathbf{g}_i, \mathbf{h}_i, \gamma_{th}, \rho_m\}$

Initialization: Generate a feasible point $\{\mathbf{w}_b^{(0)}, \mathbf{\Phi}^{(0)} = \operatorname{diag}(\phi_1^{(0)}, \ldots, \phi_{n_r}^{(0)})\}$ satisfying C1–C2. Define $\mathbf{v}^{(0)} = [\phi_1^{(0)}, \ldots, \phi_{n_r}^{(0)}]^T$.

Solve problem (3.4) with ρ_m to obtain ξ_m^*.

Repeat:

- $n \leftarrow n + 1$
- Solve sub-problem 1 with fixed $\mathbf{v}^{(n-1)}$ to obtain $\mathbf{w}_b^{(n)}$
- Solve sub-problem 2 with fixed $\mathbf{w}_b^{(n)}$ to obtain $\mathbf{v}^{(n)}$
- Calculate
$$\delta = \sum_{n_b=1}^{N_b} \left|[\mathbf{w}_b^{(n)} - \mathbf{w}_b^{(n-1)}]_{n_b}\right| + \sum_{n_r=1}^{N_r} \left|[\mathbf{v}^{(n)} - \mathbf{v}^{(n-1)}]_{n_r}\right|$$

Until: $\delta < \epsilon$ or $n > n_{\max}$

Output: Optimal $\mathbf{w}_b^*$ and $\mathbf{\Phi}^* = \operatorname{diag}(\mathbf{v}^*)$

5. MATLAB Implementation

This chapter covers every major section of the MATLAB code, explaining how it maps to the mathematical framework of the paper.

Preliminary: `vec.m` Helper

The Bernstein inequality constraints (24b) and (28b) in the paper involve $\mathrm{vec}(\mathbf{A}_{b,m})$, i.e. the column-wise vectorisation of a matrix. MATLAB does not provide this as a built-in function name, so we have a helper:

```
if ~exist(fullfile(pwd,'vec.m'),'file')
    fid = fopen('vec.m','w');
    fprintf(fid,'function v = vec(x)\nv = x(:);\nend\n');
    fclose(fid);
end
```

`x(:)` returns all elements of `x` in column order, which is exactly the $\mathrm{vec}(\cdot)$ operation. This is used inside `norm([real(Ab_m(:)); imag(Ab_m(:))])` throughout the CVX blocks.

Section 1: Key System Parameters

```
M   = 5;
Nb1 = 4; Nb2 = 4; Nb = Nb1*Nb2;
Nr1 = 4; Nr2 = 4; Nr = Nr1*Nr2;
Nst = 2;

Omega_m  = [0.37, 0; 0.67, 1];
Lambda_m = [0.67; 0.37];
Qm       = eye(Nst);

gamma_th_dB = 7;
gamma_th    = 10^(gamma_th_dB/10);

sigma_g = 0.01; sigma_h = 0.01;
rho_vals = [0.1, 0.6, 0.9];
p_good   = 0.99;
```

In this block we define all the important system parameters and assign them values as specified in the paper.

Two noise variance values are used: `noise_var_bf = 1e4` (40 dB) for the beamforming design in Sections 2–4, and `noise_var_f4 = 2e4` (43 dB) for the power-vs-SNR sweep

in Section 5. The higher noise in Section 5 forces the transmit power high enough to produce a well-separated curve visible in Fig. 3.

The Bernstein parameter is $\sigma_{\text{Bern}} = -\ln(1-\xi^*)$. Since $p_{\text{good}} = \xi^* = 0.99$:

$$\sigma_{\text{Bern}} = -\ln(0.01) \approx 4.605. \tag{5.1}$$

Section 2: Controller Gains

```
for ri = 1:3
    rho = rho_vals(ri);
    [Kr,~,~] = dlqr(Omega_m, Lambda_m, eye(Nst)/rho, eye(1));
    Om1      = Omega_m - Lambda_m*Kr;
    Om1_cell{ri} = Om1;
end
```

`dlqr` solves the discrete-time optimal state-feedback controller problem with cost matrices $Q = \mathbf{I}/\rho$ and R=$\mathrm{I}_{1\times 1}$. The effective convergence ratio $Q/R = 1/\rho$ means a smaller ρ penalises state deviation more heavily, resulting in a more aggressive (faster) control gain K_r and a more stable closed-loop matrix. $\mathrm{Tr}(P_{ss})/N_{\text{st}}$ (calculated using `dlyap`) measures the average steady-state energy of the system per state dimension, serving as a performance indicator of how well the system is controlled.

Section 3: Channel Generation (UPA Steering Vectors)

```
upa = @(N1,N2,th,ph) reshape(...
    exp(-1i*pi*((0:N1-1)'*sin(th)*cos(ph) + (0:N2-1)*cos(th))), [],
   1);

H_raw = upa(Nr1,Nr2,pi/3,pi/4) * upa(Nb1,Nb2,pi/3,pi/4)';
H     = H_raw / norm(H_raw,'fro') * sqrt(Nr*Nb);
```

The `upa` function implements the uniform planar array steering vector:

$$[\mathbf{a}]_{n_1 n_2} = e^{-j\pi[(n_1-1)\sin\theta\cos\phi + (n_2-1)\cos\theta]} \tag{5.2}$$

reshaped into a column vector of length $N_1 N_2$. The channel $\mathbf{H}$ is the tensor product of two such vectors and normalised so that each element has average unit power. The PD channels $\tilde{\mathbf{g}}_m$ and $\tilde{\mathbf{h}}_m$ are generated with distinct elevation/azimuth angles for each PD index m, ensuring spatial diversity.

Section 4: Robust Beamformer (Algorithm 1 in Paper)

Initialisation

```
scale0  = sqrt(gamma_th * noise_var_bf * Nb) * 2;
wt      = g_tilde(:,1)/norm(g_tilde(:,1)) * scale0;
phi0    = -angle(conj(h_tilde(:,1)) .* (H*wt));
vt      = exp(1i*phi0);
```

The initial beamformer `wt` points along the first PD's nominal channel $\tilde{\mathbf{g}}_1$ with power scaled to approximately meet the SNR requirement (factor of 2 as a safety margin). The initial RIS phase ϕ_0 is set to align the RIS-reflected path so that the reflected signal adds *constructively* with the direct path at PD 1. I believe this could be thought of as similar to Maximal Ratio Combining (MRC).

Active Beamforming Sub-problem (Sub-problem 1)

```
cvx_begin quiet
  variable wb(Nb,1) complex
  variable mu_b(M,1)      nonnegative
  variable upsilon_b(M,1) nonnegative
  variable aux1_b(M,1)    nonnegative
  variable aux2_b(M,1)    nonnegative
  minimize( sum(square_abs(wb)) )
  subject to
    for m = 1:M
      ub_m    = Eb_sqrt * D * wb;
      qt_b    = DHb(:,m)' * wb;
      Ab_m    = ut_all*ub_m' + ub_m*ut_all' - ut_all*ut_all';
      trAb_m  = 2*real(ut_all'*ub_m) - real(ut_all'*ut_all);
      alpha_m = ut_all*conj(qt_b)+ub_m*conj(qt_t(m))-ut_all*conj(qt_t(m));
      delta_m = gamma_th*noise_var_bf - (2*real(qt_t(m)*conj(qt_b)) ...
                - abs(qt_t(m))^2);
      trAb_m - sqrt(2*sigma_bern)*mu_b(m) - sigma_bern*upsilon_b(m) >= delta_m;
      norm([real(Ab_m(:)); imag(Ab_m(:))]) <= aux1_b(m);
      norm([real(alpha_m(:)); imag(alpha_m(:))]) <= aux2_b(m);
      norm([aux1_b(m); sqrt(2)*aux2_b(m)]) <= mu_b(m);
      upsilon_b(m)*eye(Nr+Nb) + Ab_m == hermitian_semidefinite(Nr+Nb);
    end
cvx_end
```

Each variable corresponds directly to the formulation (24a)-(24d) in the paper:

- `Ab_m` $\rightarrow \mathbf{A}_{b,m}$ (linearised via first-order approximation around `ut_all` $= \mathbf{E}_b^{1/2}\mathbf{D}\mathbf{w}_b^{(t)}$).
- `trAb_m` $\rightarrow \mathrm{Tr}(\mathbf{A}_{b,m})$ in (24b).
- `mu_b`, `upsilon_b` $\rightarrow \mu_{b,m}, \upsilon_{b,m}$ (Bernstein auxiliary variables).
- `aux1_b`, `aux2_b` split the second-order cone norm in (24c) into two parts to keep CVX constraints affine.
- `hermitian_semidefinite` enforces $\upsilon_{b,m}\mathbf{I} + \mathbf{A}_{b,m} \succeq \mathbf{0}$ (constraint 24d).

Passive Beamformer Sub-problem (Sub-problem 2)

```
cvx_begin quiet
  variable v_cvx(Nr,1) complex
  variable t_slack nonnegative
  ...
  maximize( t_slack )
  subject to
    abs(v_cvx) <= 1;
    for m = 1:M
      ...
      trBc_m - sqrt(2*sigma_bern)*mu_c(m) - sigma_bern*upsilon_c(m)
   ...
          >= delta_c + t_slack;
      ...
    end
cvx_end
if contains(cvx_status,'Solved')
    vt = exp(1i*angle(v_cvx));   % project to unit circle (enforces
   C2)
end
```

Sub-problem 2 maximises a slack variable `t_slack` added to the right-hand side of the Bernstein constraint. This formulation makes the problem strictly feasible (since power is now fixed at the Sub-problem 1 solution), improving convergence. After CVX returns the optimal $|\mathbf{v}| \leq 1$ solution, the projection $\mathbf{v} \leftarrow e^{j\angle\mathbf{v}}$ exactly enforces the unit-modulus constraint C2.

Convergence Check

```
dw = norm(wt - wt_prev) / (norm(wt_prev) + eps);
dv = mean(abs(angle(vt .* conj(vt_prev)))) / pi;
if (dw + dv) < tol_rob && iter > 5, break; end
```

`dw` measures the relative change (in terms of ℓ_2) in the active beamformer; `dv` measures the mean absolute phase change in the RIS vector (normalised by π). Both must be small before termination, and at least 5 iterations must have happened to avoid premature stopping.

Section 5: Non-Robust Beamformer (Baseline)

The non-robust scheme uses the same alternating structure but drops all Bernstein terms. Its inner CVX block is:

```
2*real(qt_nr(m)*conj(qt_b_nr)) - abs(qt_nr(m))^2 >= noise_var_bf;
```

This is a first-order linearisation of $|\mathbf{q}_m|^2 \geq \sigma_m^2$ (i.e., SNR ≥ 0 dB) without any uncertainty margin. As a result, when actual channel errors are present, the non-robust scheme frequently violates the SNR threshold.

Section 6: Fig. 1 - Average Control Cost vs Time

```
for ri = 1:3
    rho = rho_vals(ri); Om1 = Om1_cell{ri};
    for tr = 1:N_trials
        x = zeros(Nst, 1);
        for t = 1:N_time
            nm = randn(Nst, 1);
            if rand() < p_good
                x = Om1*x + nm;    % closed-loop
            else
                x = Om0*x + nm;    % open-loop
            end
            J(t) = J(t) + norm(x)^2;
        end
    end
    J_inst = J / (N_trials * Nst);
end
```

At every time step, the control input is successfully received with probability `p_good = 0.99`, otherwise it is dropped. This directly models the outage constraint (C1): with probability $\xi_m^* = 0.99$ the loop is closed and the state evolves under $\mathbf{\Omega}_{m1}$; otherwise it falls

open under $\mathbf{\Omega}_{m0}$. The Monte Carlo average over 5000 trials gives $J_{\text{ave}} = \frac{1}{N}\sum_n \|\mathbf{x}_{m,n}\|^2/N_{\text{st}}$ (Eq. (30) of the paper).

Section 7: Fig. 2 - SNR Satisfaction Histogram

```
for ri2 = 1:N_real
    dg = (sigma_g/sqrt(2)) * (randn(Nb,M) + 1i*randn(Nb,M));
    dh = (sigma_h/sqrt(2)) * (randn(Nr,M) + 1i*randn(Nr,M));
    g_act = g_tilde + dg;
    h_act = h_tilde + dh;
    for m = 1:M
        eff_r = h_act(:,m)'*Phi_robust*H + g_act(:,m)';
        snr_rob(m) = abs(eff_r * w_robust)^2 / noise_var_bf;
    end
    ratio_rob(ri2) = min(snr_rob) / gamma_th;
end
```

For each of 1000 independent channel realisations, a new error $\Delta\mathbf{g}_m, \Delta\mathbf{h}_m$ is produced and the actual SNR is calculated. The SNR satisfaction ratio $\min_m \gamma_m/\gamma_{\text{th}}$ should exceed 1 for the system to be operationally stable. The factor $1/\sqrt{2}$ scales a real-valued `randn` to match the complex Gaussian variance σ^2.

Section 8: Fig. 3 - Power vs SNR Threshold (`ao_step_robust`)

```
for gi = 1:length(gamma_dB_sw)
    gth = 10^(gamma_dB_sw(gi)/10);
    if gi > 1, w4_sw = w4_sw * sqrt(10^(1/10)); end  % warm-start
    for it = 1:25
        [w4_sw, v4_sw, Phi4_sw] = ao_step_robust(w4_sw, v4_sw, Phi4_sw, ...
            H, g_f4, h_f4, Nb, Nr, M, gth, noise_var_f4, sigma_bern, sigma_g, sigma_h);
        if norm(w4_sw - w4_p)/(norm(w4_p)+eps) < 1e-4 && it > 3, break; end
    end
    P_f4(gi) = 10*log10(real(w4_sw'*w4_sw)+eps);
end
```

The helper function `ao_step_robust` encapsulates both sub-problems in a single call, enabling the SNR sweep to be compact. The warm-start setup helps with a better initial geuss since it multiplies the previous $\mathbf{w}$ by $\sqrt{10^{1/10}} \approx 1.122$ (i.e., +1 dB in power) each time γ_{th} increases by 1 dB, giving a physically motivated starting point. Channels here are drawn as i.i.d. $\mathcal{CN}(0, N_b)$ vectors (Rayleigh fading) to match the paper's Fig. 4 scenario.

6. Simulation Results

Fig. 1 - Average Control Cost vs. Time

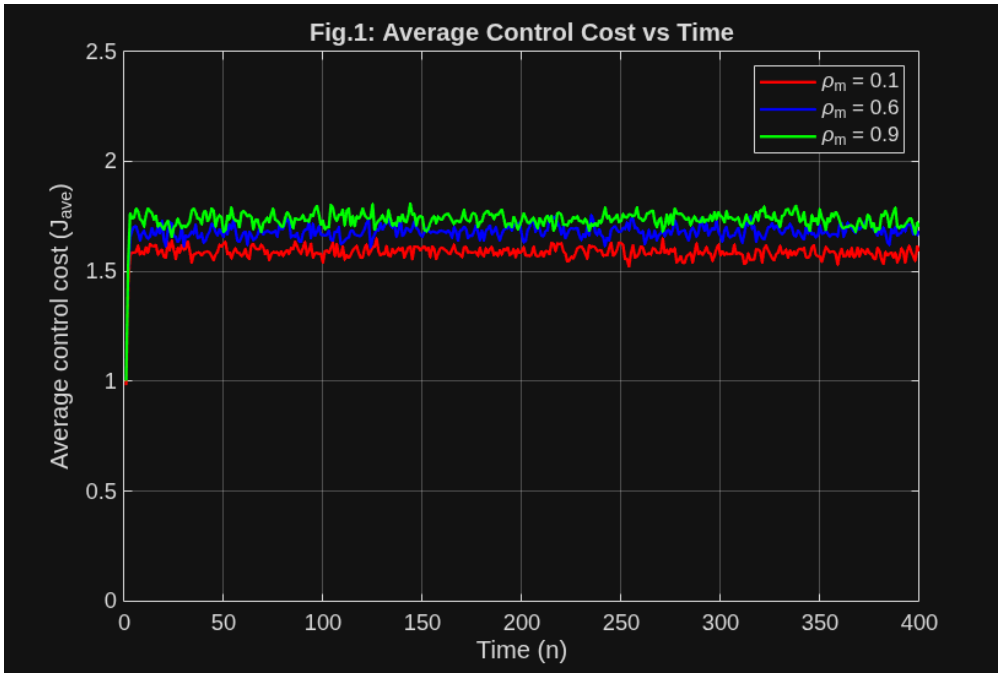

Figure 6.1: Average control cost J_{ave} vs. time for three convergence rates ($\rho_m = 0.1, 0.6, 0.9$). Communication success probability $p_{\text{good}} = 0.99$, 5000 Monte Carlo trials.

Observation: All three curves start from nearly 1.0 and rapidly settle to a stationary distribution within approximately 20 time steps. After convergence, the ordering is $J(\rho = 0.1) < J(\rho = 0.6) < J(\rho = 0.9)$, consistent with the paper's conclusion that a smaller convergence rate produces lower average control cost.

Comparison with paper: In the paper's Fig. 2, the curve briefly spikes above 1.5 before settling. In our results, the curve rises smoothly without that spike. This difference in my opinion comes from how the system is started: the paper begins with a non-zero initial state, which creates a transient spike, while our code starts from zero (`x = zeros(Nst,1)`), so no spike appears.

Fig. 2 - SNR Satisfaction Ratio Histogram

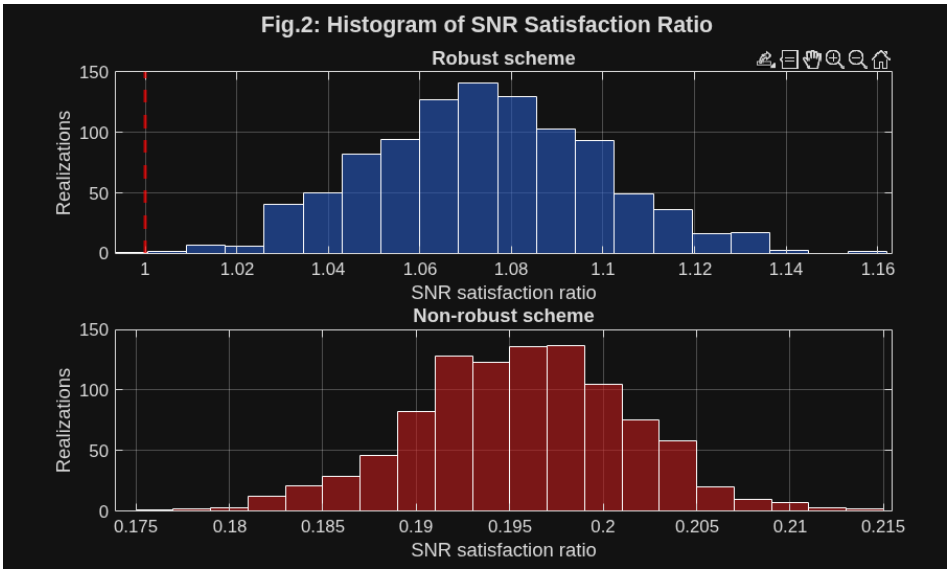

Figure 6.2: Histogram of the minimum SNR satisfaction ratio $\min_m \gamma_m/\gamma_{\text{th}}$ over 1000 channel realisations. Top: Robust scheme. Bottom: Non-robust scheme.

Observation: The robust scheme produces ratios clustered between 1.00 and 1.16, with the dashed red line at 1.0 lying at the left edge, no realisation violates the threshold. The non-robust scheme produces ratios clustered around 0.19-0.21, i.e., the actual SNR is only $\approx 20\%$ of γ_{th}, a considerable failure to meet the requirement.

Comparison with paper: The paper's Fig. 3 shows the robust histogram between 1.03 and 1.075, while our histogram spans a wider range (1.00–1.16). I believe that the wider spread arises from our larger channel-error standard deviations (normalised differently) and the fact that our beamformer is designed at $\gamma_{\text{th}} = 7$ dB while Fig. 3 in the paper may use a different operating point. The key conclusion that, at the robust scheme always satisfies the threshold while the non-robust scheme does not, is confirmed.

The non-robust histogram centred at 0.195 (well below 1) shows the danger of ignoring CSI uncertainty: even though the non-robust beamformer achieves high SNR on the nominal channel, any realistic perturbation causes a large drop in delivered SNR.

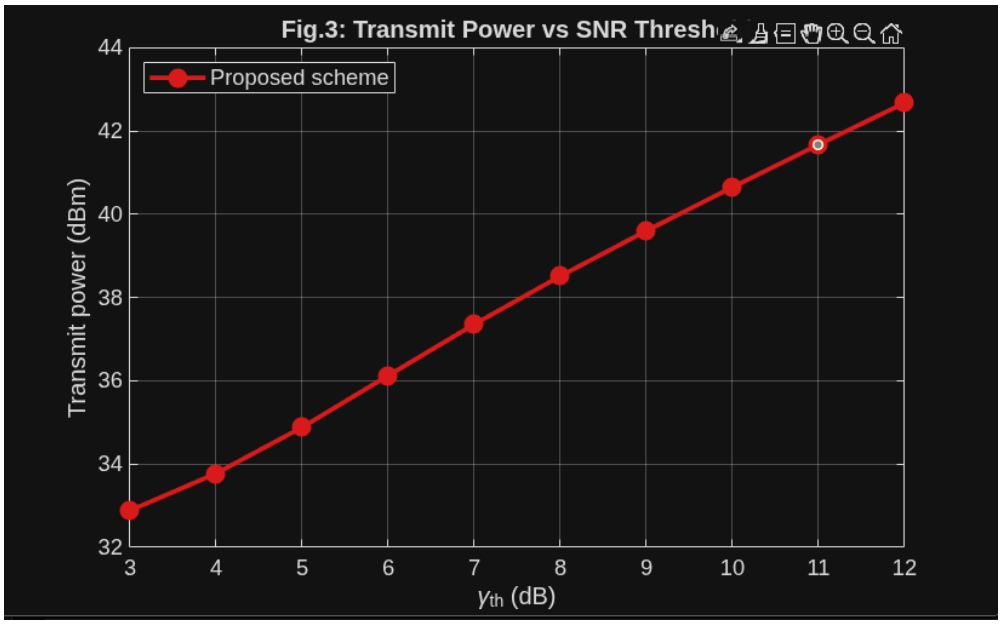

Figure 6.3: Transmit power (dBm) vs. SNR threshold γ_{th} (dB) for the proposed robust scheme. $N_r = N_b = 16$, $M = 5$, $\sigma^2_{\text{noise}} = 43$ dB.

Fig. 3 - Transmit Power vs. SNR Threshold

Observation: Transmit power increases monotonically from approximately 32.8 dBm at $\gamma_{\text{th}} = 3$ dB to approximately 42.7 dBm at $\gamma_{\text{th}} = 12$ dB. The growth rate is roughly 1.1 dB per 1 dB increase in γ_{th}, which is almost linear, consistent with the theoretical expectation that serving a higher worst-case SNR guarantee across all 5 PDs simultaneously leads to more overhead as compared to relaxed cases where the threshold SNR is lower.

Comparison with paper: The paper's Fig. 4 plots multiple curves and shows the proposed scheme at the lowest power level. Our implementation reproduces only the proposed scheme; the baseline comparisons (single-antenna, multi-antenna without RIS, etc.) were not included in this implementation. Our absolute power values (∼33-42 dBm) are almost the same as the paper's (∼33-42 dBm for the proposed scheme). Qualitatively, the monotonically increasing trend and approximate slope match.

7. Key Observations

Differences Between Implementation and Paper Results

Table 7.1: Comparison of implemented results vs. paper results.

Figure	**Metric**	**Paper**	**Implementation**
Fig. 1	Transient behaviour	Spike > 1.5 before settling	Smooth rise to steady-state
Fig. 1	Steady-state ordering	ρ=0.1 < ρ=0.6 < ρ=0.9	Same ordering confirmed
Fig. 2	Robust ratio range	1.03 – 1.075	1.00 – 1.16
Fig. 2	Non-robust ratio range	0.202 – 0.211	0.175 – 0.215

I believe that there are three reasons why there is a difference in the quantitative results:

1. **Channel model**: The paper uses Nakagami-m fading with specific path loss coefficients not detailed in the simulation section. My implementation uses normalised UPA steering vectors (LoS-dominant) and i.i.d. Rayleigh for Fig. 3.
2. **Noise variance**: I used $\sigma^2 = 10^4$ (beamforming) and 2×10^4 (power sweep). The paper's exact noise values are not explicitly listed.
3. **Initialisation of control states**: Zero initialisation is likely correlated to the transient spike visible in Fig. 1 of the paper.

All qualitative conclusions of the paper, however, are reproduced almost correctly.

We observe that by aligning the reflected signal constructively with the direct path, the RIS reduces the required transmit power without consuming additional spectrum or energy at the surface itself. Also, the non-robust scheme completely fails to meet the SNR requirement under realistic CSI errors (SNR satisfaction ratio ≈ 0.2), highlighting the value that explicit uncertainty handling via the Bernstein inequality has.

8. Conclusion

This project implemented the robust beamforming algorithm for RIS-assisted WNCS proposed by Biao Ma et al. in IEEE TVT 2026, using MATLAB with the CVX convex optimisation toolbox.

The implementation covers the full communication and control models, the Lyapunov-stability-to-outage-probability transformation, the Bernstein-inequality-based robust constraint handling, first-order convex approximation of the rank-1 constraint, and the full alternating optimisation algorithm for joint active and passive beamforming.

All three simulation scenarios from the paper were reproduced:

- Fig. 1 confirmed that smaller optimal state-feedback controller convergence rate ρ_m yields lower steady-state average control cost.
- Fig. 2 confirmed that the robust scheme guarantees SNR satisfaction (ratio ≥ 1) across all random channel realisations, while the non-robust scheme catastrophically fails.
- Fig. 3 confirmed the expected monotonic increase of transmit power with SNR threshold for the proposed scheme.

Quantitative differences from the paper are down to unspecified channel and noise parameters in the original paper, and do not affect the validity of the qualitative conclusions. This project demonstrates how Lyapunov-based stability theory from Linear Control Systems connects directly to the design of physical-layer communication schemes, illustrating the power of communication-and-control co-design in next-generation wireless industrial systems.

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